Zhe (Barry) Chen

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WORK EXPERIENCE

2025.01–Present **The Hong Kong Polytechnic University** **Postdoctoral Fellow**
Supervisor: Prof. Fu Xiao

2020.07–2021.07 **University of Shanghai for Science and Technology** **Research Assistant**
Supervisor: Prof. Yongbao Chen

EDUCATION

2021.09–2024.12 **The Hong Kong Polytechnic University** **PhD**
Supervisor: Prof. Fu Xiao

2017.09–2020.04 **Tongji University** **MEng**
Supervisor: Prof. Peng Xu

2013.09–2017.06 **Nanjing University of Aeronautics and Astronautics** **BEng**

AWARDS & HONORS

2026 **Best Oral Presentation Award and Best Paper Nomination**, The International Conference of Building and Simulation (BAS 2026)

2025 **Champion (Ranking 1/610, AU$10,000)**, FlexTrack Challenge—Detecting Energy Flexibility in Buildings, AIcrowd, New South Wales Government, and University of Wollongong

2025 **PolyU FCE Awards for Outstanding PhD Theses**, The Hong Kong Polytechnic University (Top 2.5%; only recipient from the department)

2024 **PolyU Distinguished Postdoctoral Fellowship (HK$841,040)**, The Hong Kong Polytechnic University; finalist for the RGC Junior Research Fellow Scheme (the only nominee from the department)

2024 **Elsevier Highly Cited Paper Award**, the most cited paper published in ***Advances in Applied Energy*** (IF: 18.3, Q1) in 2023

2023 **Champion (CN¥100,000)**, First Greater Bay Area Data Application Innovation Competition, Development and Reform Commission of Shenzhen Municipality

2023 **Excellent Control Award**, 21st MDV Central Air Conditioning Design and Application Competition, Midea Building Technologies

2023 **Champion (Ranking 1/331, CN¥20,000)**, AI Master Cup Algorithm Competition, Schneider Electric and Baidu PaddlePaddle

2023 **ESI Hot Paper and Highly Cited Paper**, corresponding author paper titled "Physical energy and data-driven models in building energy prediction: A review"

2023 **Champion (Ranking 1/513, CN¥20,000)**, Smart Energy System Algorithm Competition, UNiLAB and Alibaba Tianchi

2022 **Paper nominated for the Eni Award 2023**, first author paper titled "Multi-objective residential load scheduling approach for demand response in smart grid"

2022 **Champion**, i-Space: VRAR Contest 2021/22, project titled "AR-enabled smart building energy management platform", The Hong Kong Polytechnic University

2022 **Silver Award**, Global AI Challenge for Building E&M Facilities, Electrical and Mechanical Services Department

2021 **PolyU Presidential PhD Fellowship (HK$1,203,800)**, The Hong Kong Polytechnic University

PUBLICATIONS

Citations: 2,300; h-index: 18; hot paper: 1; highly cited paper: 2

Journal Papers

1. **Chen Z**, Xiao F*, Guo F, Yan J. Interpretable machine learning for building energy management: A state-of-the-art review. *Advances in Applied Energy* 2023;9:100123. (IF: 18.3, Q1, Elsevier Highly Cited Paper Award)
2. **Chen Z**, Xiao F*, Guo F. Similarity learning-based fault detection and diagnosis in building HVAC systems with limited labeled data. *Renewable and Sustainable Energy Reviews* 2023;185:113612. (IF: 18.0, Q1)
3. **Chen Z**, Chen Y*, He R, Liu J, Gao M, Zhang L. Multi-objective residential load scheduling approach for demand response in smart grid. *Sustainable Cities and Society* 2022;76:103530. (IF: 13.3, Q1, Nominated for the Eni Award 2023)
4. **Chen Z**, Xiao F*, Chen Y. Multi-objective online optimization of building energy systems for improved control smoothness and efficiency. *Automation in Construction* 2026;181:106604. (IF: 12.6, Q1)
5. **Chen Z**, Zhang J, Xiao F*, Xu K, Chen Y. Development of a probabilistic cooling load prediction-based robust chiller sequencing strategy and its real-world implementation. *Applied Energy* 2025;382:125213. (IF: 12.2, Q1)
6. **Chen Z**, Zhang J, Xiao F*, Madsen H, Xu K. Probabilistic machine learning for enhanced chiller sequencing: A risk-based control strategy. *Energy and Built Environment* 2025;6:783–95. (IF: 10.4, Q1)
7. **Chen Z**, Xiao F*, Xiao Z, Chen Y. Bridging the gap between data-driven baselines and energy saving uncertainty for building retrofit. *Energy* 2025;340:139292. (IF: 10.1, Q1)
8. **Chen Z**, Zhang J, Xiao F*, Xu K, Chen Y. Physically consistent data-driven optimal sequencing strategy for variable speed pumps in large building chiller plants. *Building and Environment* 2025;281:113177. (IF: 8.4, Q1)
9. **Chen Z**, Chen Y*, Xiao T, Wang H, Hou P. A novel short-term load forecasting framework based on time-series clustering and early classification algorithm. *Energy and Buildings*

2021;251:111375. (IF: 8.0, Q1)

10. **Chen Z**, Xu P*, Feng F, Qiao Y, Luo W. Data mining algorithm and framework for identifying HVAC control strategies in large commercial buildings. ***Building Simulation*** 2021;14:63–74. (IF: 6.9, Q1)
11. Chen Y, **Chen Z***. Short-term load forecasting for multiple buildings: A length sensitivity-based approach. ***Energy Reports*** 2022;8:14274–88. (IF: 6.6, Q2, Corresponding author)
12. Chen Y*, Guo M, Chen Z, **Chen Z***, Ji Y. Physical energy and data-driven models in building energy prediction: A review. ***Energy Reports*** 2022;8:2656–71. (IF: 6.6, Q2, Corresponding author, ESI Hot Paper, ESI Highly Cited Paper)
13. Xu K, **Chen Z**, Xiao F*, Zhang J, Zhang H, Ma T. Semantic model-based large-scale deployment of AI-driven building management applications. ***Automation in Construction*** 2024;165:105579. (IF: 12.6, Q1)
14. Tang H, **Chen Z**, Li H, Wang S*. Risk-aware optimal dispatch of resource aggregators integrating NGBoost-based probabilistic renewable forecasting and bi-level building flexibility engagements. ***Engineering*** 2025;53:76–89. (IF: 12.2, Q1)
15. Chen Y, **Chen Z**, Xu P*, Li W, Sha H, Yang Z, et al. Quantification of electricity flexibility in demand response: Office building case study. ***Energy*** 2019;188:116054. (IF: 10.1, Q1)
16. Chen Y*, **Chen Z**, Yuan X, Su L, Li K*. Optimal control strategies for demand response in buildings under penetration of renewable energy. ***Buildings*** 2022;12:371. (IF: 3.4, Q2)
17. Chen Y, **Chen Z**, Ding Y, Du Y*, Li B, Liu J, et al. Towards transformer winding deformation fault diagnosis under data scarcity: Integrating frequency response analysis with metric learning. ***IEEE Transactions on Dielectrics and Electrical Insulation*** 2026. (IF: 3.3, Q2)
18. Chen Y, Xu P*, **Chen Z**, Wang H, Sha H, Ji Y, et al. Experimental investigation of demand response potential of buildings: Combined passive thermal mass and active storage. ***Applied Energy*** 2020;280:115956. (IF: 12.2, Q1)
19. Song Z, Xiao F*, **Chen Z**, Madsen H. Probabilistic ultra-short-term solar photovoltaic power forecasting using natural gradient boosting with attention-enhanced neural networks. ***Energy and AI*** 2025;20:100496. (IF: 9.7, Q1)
20. Chen Y*, Yang Q*, **Chen Z**, Yan C, Zeng S, Dai M. Physics-informed neural networks for building thermal modeling and demand response control. ***Building and Environment*** 2023;234:110149. (IF: 8.4, Q1)
21. Chen Y*, Wang H*, **Chen Z**. Sensitivity analysis of physical regularization in physics-informed neural networks (PINNs) of building thermal modeling. ***Building and Environment*** 2025;273:112693. (IF: 8.4, Q1)
22. Xiao Z, Zhang J, **Chen Z**, Xiao F*. Probabilistic building energy savings verification via triplet network-based adaptive reference day selection. ***Energy and Buildings*** 2026;364:117664. (IF: 8.0, Q1)
23. Chen Y*, Chen Z, **Chen Z**, Yuan X. Dynamic modeling of solar-assisted ground source heat pump using Modelica. ***Applied Thermal Engineering*** 2021;196:117324. (IF: 7.5, Q1)
24. Yuan H, Chen Y*, **Chen Z**. Investigation on electricity flexibility and demand-response strategies for grid-interactive buildings. ***Buildings*** 2025;15:4368. (IF: 3.4, Q2)

25. Xiao T, Xu P*, Ding R, **Chen Z**. An interpretable method for identifying mislabeled commercial building based on temporal feature extraction and ensemble classifier. ***Sustainable Cities and Society*** 2022;78:103635. (IF: 13.3, Q1)
26. Chu Y, Xu P*, Li M, **Chen Z**, Chen Z, Chen Y, et al. Short-term metropolitan-scale electric load forecasting based on load decomposition and ensemble algorithms. ***Energy and Buildings*** 2020;225:110343. (IF: 8.0, Q1)
27. Xiao Z, Zhang J*, Xiao F*, **Chen Z**, Xu K, So PM, et al. An AI-enabled optimal control strategy utilizing dual-horizon load predictions for large building cooling systems and its cloud-based implementation. ***Energy and Buildings*** 2025;330:115352. (IF: 8.0, Q1)
28. Yan D*, Dong B, O'Neill Z, **Chen Z**, Wang X, Jiang Z, et al. Whole-process AI in building applications: A framework of integrating AI with domain knowledge. ***Building Simulation*** 2026. (IF: 6.9, Q1)
29. Han C, Chen Y*, Wang H, Zhan S, **Chen Z**. Development and validation of a data quality assessment framework for machine learning-based building load prediction. ***Applied Energy*** 2026;415:127925. (IF: 12.2, Q1)
30. Liu J, Chen Y*, Yang J, Wu W, **Chen Z**. A meta energy model-based transfer learning methodology in cooling load predictions: Office building case study. ***Energy*** 2026;360:141592. (IF: 10.1, Q1)
31. Sha H, Xu P*, Hu C, Li Z, Chen Y, **Chen Z**. A simplified HVAC energy prediction method based on degree-day. ***Sustainable Cities and Society*** 2019;51:101698. (IF: 13.3, Q1)
32. Li A, Zhang J, Xiao F*, Fan C, Yu Y, **Chen Z**. Design information-assisted graph neural network for modeling central air conditioning systems. ***Advanced Engineering Informatics*** 2024;60:102379. (IF: 11.5, Q1)
33. Gu J, Xu P*, Pang Z, Chen Y, Ji Y, **Chen Z**. Extracting typical occupancy data of different buildings from mobile positioning data. ***Energy and Buildings*** 2018;180:135–45. (IF: 8.0, Q1)

Conference Papers and Presentations

1. **Chen Z**, Xiao F*. Fast flexibility quantification and resilience evaluation for building air-conditioning systems. *The International Conference of Building and Simulation* **(BAS 2026)**, June 2026, Hong Kong, China.
2. **Chen Z**, Xiao F*. A novel demand response detection and capacity prediction framework. *Applied Energy Symposium* **(AI×Energy 2026)**, May 2026, Ningbo, China.
3. **Chen Z**, Xiao F*. Reliably estimating energy savings using data-driven models as the baseline. *The International Conference of Building and Simulation* **(BAS 2025)**, July 2025, Borlänge, Sweden.
4. **Chen Z**, Xiao F*. Robust comparison of optimization algorithms for optimal chiller loading. *International Conference on Sustainable Development in Building and Environment* **(SuDBE 2023)**, Aug. 2023, Espoo, Finland.
5. **Chen Z**, Xiao F*. Similarity learning-based FDD in building HVAC systems. *The 3rd International Conference for Global Chinese Academia on Energy and Built Environment* **(CEBE 2023)**, July 2023, Shanghai, China.

6. **Chen Z**, Chen Y*, Mohtaram S. Quantifying HVAC electrical flexibility from building thermal mass: A case study of the DOE reference building. *The 16th ROOMVENT Conference* **(ROOMVENT 2022)**, Sept. 2022, Xi'an, China.
7. **Chen Z**, Xiao F*. Interpretable machine learning for AHU fault detection and diagnosis: Rule extraction from black-box models. *The 14th International Conference on Applied Energy* **(ICAE 2022)**, Aug. 2022, Bochum, Germany.
8. **Chen Z**, Chen Y*. A multi-objective optimization-based load scheduling approach for demand response in smart grid. *The 2nd International Conference for Global Chinese Academia on Energy and Built Environment* **(CEBE 2021)**, July 2021, Chengdu, China.
9. **Chen Z**, Xu P*, Chen Y. A peer-to-peer electricity system and its simulation. *The 4th Asia Conference of International Building Performance Simulation Association* **(ASIM 2018)**, Dec. 2018, Hong Kong, China.

Conference Session Chairs

1. *The 2nd International Conference on Digital Intelligence for Energy Systems* **(ICDIES 2026)**, Oct. 2026, Wollongong, Australia.
2. *Applied Energy Symposium* **(AI×Energy 2026)**, May 2026, Ningbo, China.

Invited Talks

1. Demand response detection and capacity prediction using ensemble learning. *IEEE International Conference on Energy Technologies for Future Grids* **(IEEE ETFG 2025)**
   Dec. 2025, Wollongong, Australia. (Invited as the winner of the FlexTrack Challenge)
2. Embarking on your research journey: My experience at PolyU BEEE. *BEEE Research Postgraduate Student Orientation, The Hong Kong Polytechnic University*
   Sept. 2025, Hong Kong, China. (Invited as the PhD graduate representative)

PATENTS

1. A chiller control method, apparatus, equipment, and storage medium, CN118189539B.
2. A method, apparatus, device, and program product for determining the number of operating water pumps, CN122041288A.
3. A demand response control method for multi-energy residential buildings, CN112968445B.
4. An automatic generation method for schematic diagrams of HVAC water systems based on topological layered abstraction, CN111797485B.
5. A path optimization method for water systems in air conditioning systems, CN112241564B.
6. An optimization algorithm for pipe shafts, CN112241563B.
7. An automatic equipment selection algorithm for HVAC systems, CN113268796B.

RESEARCH PROJECTS

2025–2027	**Development of smart and scalable data-driven management technologies for building HVAC systems towards carbon neutrality** PolyU Distinguished Postdoctoral Fellowship Scheme, P0053872, HK$841,040 Role: Fellowship Awardee
2022–2024	**Smart data-driven building management framework for environmental and sustainability applications** Innovation and Technology Fund of Hong Kong, ITP/002/22LP, HK$7,300,000 Role: Lead researcher for demonstration of data-driven building management technologies in large buildings
2021–2024	**Development of smart energy management technologies for complex building energy systems in high-density cities** National Key R&D Program of China, 2021YFE0107400, CN¥4,120,000 Role: Lead researcher for data-driven modelling and optimization for building energy systems
2017–2021	**Distributed integrated energy system joint simulation technologies** National Key R&D Program of China, 2017YFB0903404, CN¥3,680,000 Role: Lead researcher for distributed integrated energy system modeling

ACADEMIC SERVICE

Youth Editor	***Building Simulation*** (IF: 6.9, Q1) ***Carbon Neutrality*** (IF: 11.2, Q1)
Guest Editor	***Buildings*** (ISSN 2075-5309), Special Issue: "Smart and Sustainable Buildings: Advancing towards Net-Zero and Intelligent Control"
Peer Reviewer	*Applied Energy, Automation in Construction, Energy Conversion and Management, Journal of Cleaner Production, Advanced Engineering Informatics, Engineering Applications of Artificial Intelligence, Computers in Industry, Energy, Building and Environment, Applied Thermal Engineering, Energy and Buildings, Building Simulation, iScience*
Membership	International Building Performance Simulation Association International Society of Energy and Built Environment

TEACHING EXPERIENCE

Advising	BSE3716 & BSE4726 (Capstone Research Project), 2025–2026	6 undergraduates
	BSE3713 & BSE4725 (Capstone Research Project), 2023–2024	2 undergraduates